Kubernetes (K8s) Notes

🔶 What is Kubernetes (K8s)

Kubernetes (K8s) is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications.

It helps manage applications running in containers (like Docker) across multiple machines efficiently.

Why do we need Kubernetes in real-world DevOps

In real-world DevOps, applications must be:

- Highly available
- Scalable
- Fault-tolerant

Kubernetes helps by:

- Automatically scaling apps based on demand
- Restarting failed containers
- Managing deployments without downtime
- Handling networking between services

Without Kubernetes, managing many containers manually becomes complex and error-prone.

Real-life Restaurant Example (Container Orchestration)

Imagine a restaurant:

- **Customers** = Users
- **Orders** = Requests

- **Kitchen** = Containers
- **Chef Manager** = Kubernetes

What Kubernetes does:

- If too many customers → adds more chefs (scaling)
- If a chef fails → replaces them (self-healing)
- Distributes orders efficiently (load balancing)

Kubernetes ensures everything runs smoothly without manual intervention.

Kubernetes Architecture

1. Master Node (Control Plane)

Responsible for managing the cluster.

Components:

- API Server → Communication hub
- Scheduler → Assigns tasks to nodes
- Controller Manager → Maintains desired state

2. Worker Nodes

These are machines where applications actually run.

They contain:

- Kubelet → Communicates with master
- Container runtime → Runs containers
- Pods → Smallest unit of deployment

3. Pods

- Smallest deployable unit in Kubernetes

- Can contain one or more containers
- Share network and storage

Key Responsibilities of Kubernetes

Auto-scaling

Automatically increases or decreases the number of containers based on traffic.

Self-healing

- Restarts failed containers
- Replaces unhealthy pods

Traffic Distribution (Load Balancing)

Distributes incoming traffic evenly across containers.

Cloud-Independent Platform

Kubernetes is platform-independent and can run on:

- AWS (Amazon Web Services)
- Azure (Microsoft Azure)
- GCP (Google Cloud Platform)
- Local machines (Minikube, Kind)

This makes Kubernetes highly flexible and portable across environments.

Summary

Kubernetes is like an intelligent manager that:

- Runs your applications

- Scales them automatically
- Fixes failures
- Works across any cloud or local setup